TASK SPECIFICATION: P9-03 PHASE P9 — PRODUCTION DEPLOYMENT, CI/CD & OPERATIONS

Automated Database Backup, Disaster Recovery & Health Monitoring

Git Branch: `chore/P9-03-backup-monitoring` · Target Pull Request: chore/P9-03-backup-monitoring

Core Objective & Technical Scope:
Automated PostgreSQL backup cron jobs, verified point-in-time recovery runbook, and Prometheus/Grafana metric telemetry.

Priority Level	High	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`chore/P9-03-backup-monitoring`
Upstream Dependencies	P9-01		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Author automated pg_dump backup script with encryption and compression.
2. Implement backup rotation policy with automated pruning.
3. Perform and document complete disaster recovery restoration drill.
4. Configure Prometheus actuator metric scraping and alert rules.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `scripts/backup*`
- `docs/runbooks/backup-restore.md`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Database restoration drill proves clean recovery with zero data loss.
- Automated alerts trigger on container downtime or high resource utilization.

VERIFICATION & TEST COVERAGE STRATEGY

- Disaster recovery restoration drill
- Alert notification test